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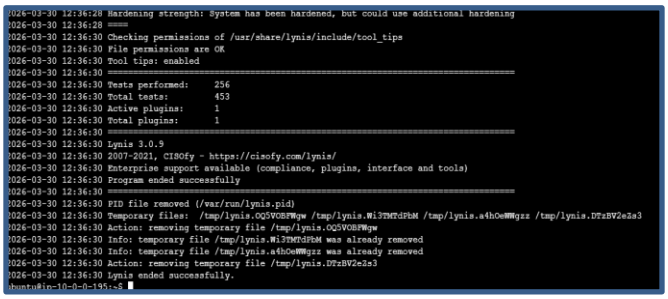
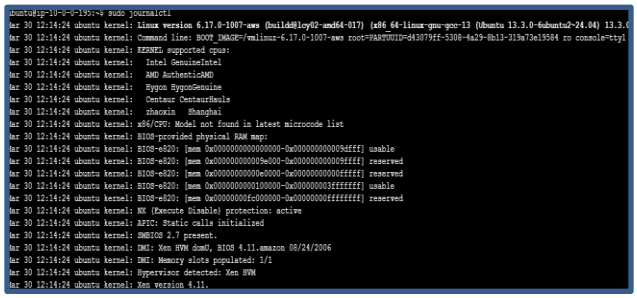
DATE: 30TH MARCH 2026

LAB ASSIGNMENT: Linux Security Monitoring and Auditing

NAME OF INSTRUCTOR: AMINU IDRIS

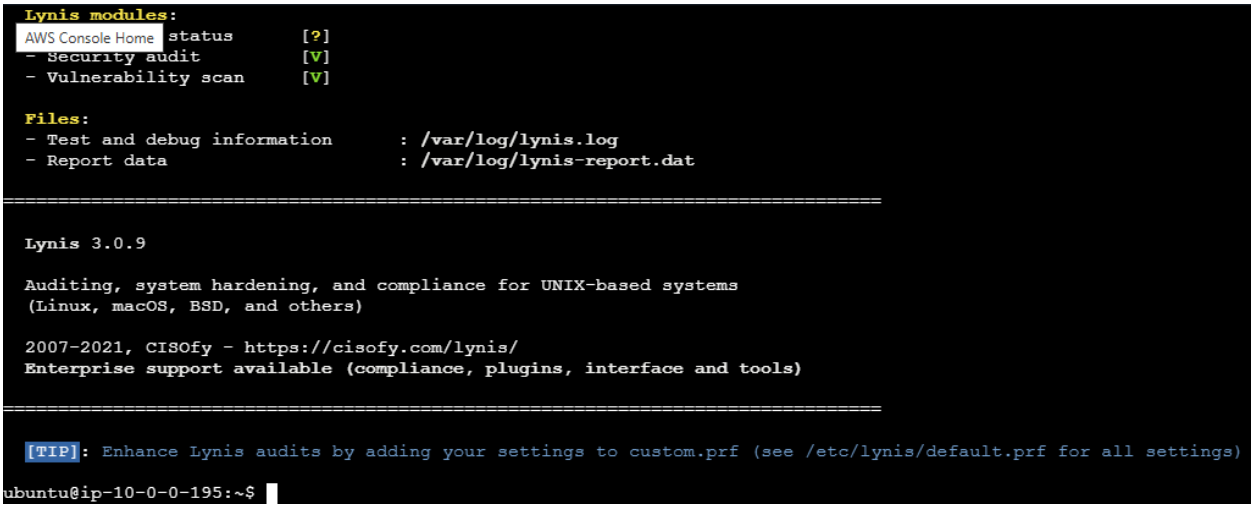
This lab was conducted on an AWS EC2 Ubuntu 24.04 LTS instance (hostname: ip-10-0-0-195) to demonstrate practical Linux security monitoring and auditing techniques. The exercises covered kernel log analysis using journalctl, SSH activity monitoring, file integrity auditing with auditd, and system hardening assessment using the Lynis security auditing tool. The findings documented here reflect real-time observations made on March 30, 2026



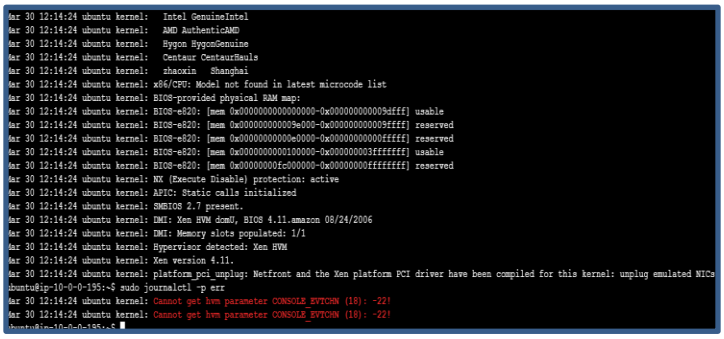


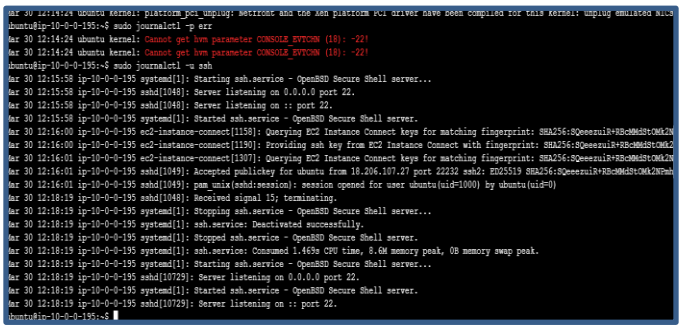
What journalctl

journalctl is the command-line interface for querying the systemd journal, which is the centralised logging system on modern Ubuntu and other systemd-based Linux distributions. It captures kernel messages, service events, authentication events, and application logs in a structured binary format. It is a critical tool for security monitoring because it allows an analyst to filter logs by priority level, time range, unit (service), or user, making it easier to isolate suspicious or notable activity without manually parsing plain-text log files

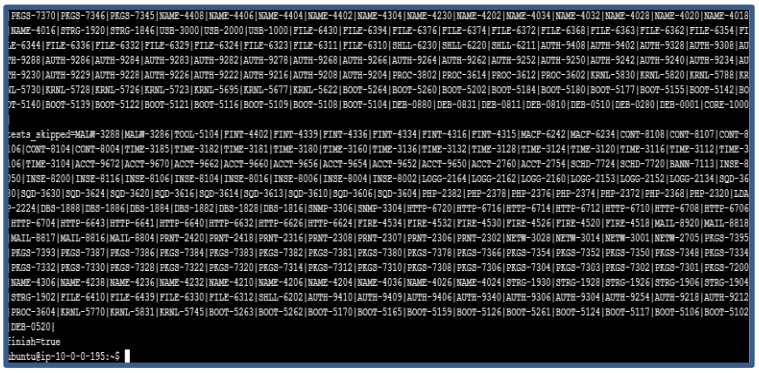


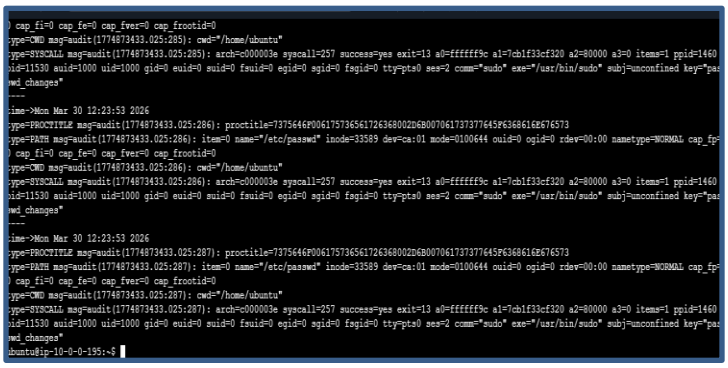
auditd is the Linux Audit Daemon — a kernel-level security auditing framework that records system calls, file accesses, user logins, and privilege escalations to a tamper-resistant audit log at /var/log/audit/audit.log. Unlike journalctl which captures service-level events, auditd operates at the syscall level, giving security analysts a granular and forensically reliable record of who did what on the system. It is a core component of compliance frameworks including PCI DSS, HIPAA, and ISO/IEC 27001, all of which require evidence of monitoring for access to sensitive files.





The SSH journal revealed a complete lifecycle of the SSH service on March 30, 2026, along with a successful authentication event.





The following audit rule was applied to monitor access to the /etc/passwd file, which stores user account information and is a common target for privilege escalation and reconnaissance

Running sudo ausearch -k passwd_changes produced multiple audit records. The following analysis covers the observed events

Interpretation: The ubuntu user (uid=1000) executed a sudo command that triggered three consecutive read attempts on /etc/passwd. The system recorded each attempt. The exit code 13 (EACCES) indicates that while the syscall was invoked, the access control layer denied full access this is normal behaviour when sudo checks credentials against PAM and /etc/passwd during authentication. This is not a suspicious event; it reflects normal sudo operation. However, the fact that auditd captured all three attempts

with full forensic detail (user, PID, path, mode, working directory) demonstrates that the audit rule is functioning correctly.